Integrating Rational Functions With Partial Fractions Answer Key

AP Calculus AB / BC   Integration Techniques

Quick Answers

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1. $-\frac{1}{x+3}+\frac{1}{x-2}$, $\log(|x-2|)-\log(|x+3|)$
2. $\frac{3}{x+2}+\frac{2}{x-1}$, $2\log(|x-1|)+3\log(|x+2|)$
3. $12\log\left(\frac{3}{2}\right)$
4. $\frac{1}{x+1}+\frac{1}{x-1}-\frac{2}{x}$, $-2\log(|x|)+\log(|x-1|)+\log(|x+1|)$
5. $6\log(2)$
6. $\log(3)$, $\log(3)$
7. $\log\left(\frac{8}{3}\right)$
8. $2\log(4)$, $\log(16)$

Curriculum

Level: AP Calculus AB / BC   Integration Techniques
FUN-6.D–FUN-6.E – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 13 tagged checks.

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1. $\displaystyle\int \frac{5}{(x-2)(x+3)}\,dx$**.**

Step 1. Part (a): the denominator is already factored into two distinct linear factors, so write $\frac{5}{(x-2)(x+3)}=\frac{A}{x-2}+\frac{B}{x+3}$ and clear denominators: $5=A(x+3)+B(x-2)$.

Step 2. Substituting the roots kills one term each time. $x=2$ gives $5=5A$, so $A=1$; $x=-3$ gives $5=-5B$, so $B=-1$.

$$\boxed{\frac{1}{x-2}-\frac{1}{x+3}}$$

Step 3. Part (b): each piece is a logarithm, since $\int\frac{dx}{x-r}=\ln|x-r|$.

$$\boxed{\ln|x-2|-\ln|x+3|+C}$$

Check by recombining: $\frac{1}{x-2}-\frac{1}{x+3}=\frac{(x+3)-(x-2)}{(x-2)(x+3)}=\frac{5}{(x-2)(x+3)}$, the original integrand. Recombining is faster than differentiating and catches a sign error immediately.

2. $\displaystyle\int \frac{5x+1}{(x-1)(x+2)}\,dx$**.**

Step 1. Part (a): what matters is the *degree*, not the shape — the numerator has degree 1 and the denominator degree 2, so no division is needed and constants over each factor still suffice. Clearing: $5x+1=A(x+2)+B(x-1)$.

Step 2. $x=1$ gives $6=3A$, so $A=2$; $x=-2$ gives $-9=-3B$, so $B=3$.

$$\boxed{\frac{2}{x-1}+\frac{3}{x+2}}$$

Step 3. Part (b): integrate term by term, keeping each constant as the coefficient of its logarithm.

$$\boxed{2\ln|x-1|+3\ln|x+2|+C}$$

Where marks are lost: the coefficients belong outside the logarithms, not inside. $2\ln|x-1|$ is $\ln(x-1)^2$, not $\ln|2x-2|$.

3. Marginal cost, $\displaystyle\int_0^1 \frac{24}{(x+1)(x+3)}\,dx$.

Step 1. Part (a): $24 = A(x+3) + B(x+1)$. At $x=-1$, $24 = 2A$ so $A = 12$; at $x=-3$, $24 = -2B$ so $B = -12$. The decomposition is $\frac{12}{x+1} - \frac{12}{x+3}$.

Step 2. Part (b): antidifferentiate and evaluate. $[12\ln|x+1| - 12\ln|x+3|]_0^1 = 12\ln\frac{2}{4} - 12\ln\frac{1}{3}$.

Step 3. Combine the logarithms: $12\left(\ln\frac{1}{2} + \ln 3\right) = 12\ln\frac{3}{2}$. $\boxed{12\ln\frac{3}{2}}$

Step 4. Part (c): the integrand is dollars per unit and x is measured in units, so the integral is in dollars — about 4.87 dollars of added cost for that increase in production.

Note on the arithmetic: writing the evaluated form as a single logarithm before computing avoids four separate rounding steps and is what "exact answer" asks for.

4. $\displaystyle\int \frac{2}{x(x-1)(x+1)}\,dx$.

Step 1. Part (a): three distinct factors need three constants: $2 = A(x-1)(x+1) + Bx(x+1) + Cx(x-1)$.

Step 2. $x=0$ gives $2 = -A$, so $A=-2$; $x=1$ gives $2 = 2B$, so $B=1$; $x=-1$ gives $2 = 2C$, so $C = 1$. $\boxed{\dfrac{-2}{x} + \dfrac{1}{x-1} + \dfrac{1}{x+1}}$

Step 3. The negative constant is not a slip: at $x=0$ the two surviving factors are $(-1)(1) = -1$, so the substitution divides 2 by -1.

Step 4. Part (b): integrate each piece. $\boxed{-2\ln|x| + \ln|x-1| + \ln|x+1| + C}$

Equivalent form: the answer can be written $\ln\left|\frac{x^2-1}{x^2}\right| + C$. Both are correct; the log-rule form is worth showing students because it makes the $x \to \infty$ behaviour visible.

5. Filtration, $\displaystyle\int_0^3 \frac{12}{(t+1)(t+3)}\,dt$.

Step 1. Part (a): $12 = A(t+3) + B(t+1)$. At $t=-1$, $12 = 2A$ so $A=6$; at $t=-3$, $12 = -2B$ so $B=-6$. So $r(t) = \frac{6}{t+1} - \frac{6}{t+3}$.

Step 2. Part (b): $[6\ln|t+1| - 6\ln|t+3|]_0^3 = 6\ln\frac{4}{6} - 6\ln\frac{1}{3} = 6\left(\ln\frac{2}{3} + \ln 3\right)$. $\boxed{6\ln 2}$

Step 3. That is about 4.16 litres in the first 3 minutes, since the rate is in litres per minute and t is in minutes.

Step 4. Part (c): both factors $t+1$ and $t+3$ increase with t, so their product increases and the quotient $12/[(t+1)(t+3)]$ decreases. The filter slows as it clogs — no graph required.

Why the answer is so clean: $\ln\frac{2}{3} + \ln 3 = \ln 2$. Simplify inside the bracket before multiplying by 6.

6. Leak, $\displaystyle\int_1^\infty \frac{2}{t(t+2)}\,dt$**.**

Step 1. Part (a): $2 = A(t+2) + Bt$. At $t = 0$, $2 = 2A$ so $A = 1$; at $t = -2$, $2 = -2B$ so $B = -1$. The antiderivative is $\ln|t| - \ln|t+2| = \ln\left|\frac{t}{t+2}\right|$.

Step 2. Part (b): on $[1, b]$ this evaluates to $\ln\frac{b}{b+2} - \ln\frac{1}{3} = \ln\frac{b}{b+2} + \ln 3$. As $b \to \infty$, $\frac{b}{b+2} \to 1$ and $\ln 1 = 0$, so only the constant survives. $\boxed{\ln 3}$

Step 3. Part (c): the limit exists and is finite, so the improper integral converges, and its value is that limit — about 1.10 kilograms of chemical in total. $\boxed{\ln 3}$

The step students skip: $\ln\frac{b}{b+2}$ must be combined into one logarithm *before* the limit. Taking $\lim(\ln b) - \lim(\ln(b+2))$ is $\infty - \infty$, which is undefined and tempts a wrong conclusion of divergence.

7. Logistic growth, $\displaystyle\int_2^4 \frac{10}{p(10-p)}\,dp$**.**

Step 1. Part (a): $10 = A(10-p) + Bp$. At $p = 0$, $10 = 10A$ so $A = 1$; at $p = 10$, $10 = 10B$ so $B = 1$. The decomposition is $\frac{1}{p} + \frac{1}{10-p}$, and both constants are positive.

Step 2. The sign question: $\int \frac{dp}{10-p} = -\ln|10-p|$, so the minus sign appears at the *integration* step, from the inner derivative -1, not in the constant B.

Step 3. Part (b): $[\ln|p| - \ln|10-p|]_2^4 = \ln\frac{4}{6} - \ln\frac{2}{8} = \ln\frac{2}{3} - \ln\frac{1}{4}$. $\boxed{\ln\frac{8}{3}}$

Step 4. Part (c): the growth rate $\frac{p(10-p)}{10}$ is largest at $p = 5$, so the stretch from 4 to 6 sits astride the fastest growth while the stretch from 2 to 4 does not. The same rise in population therefore takes less time. (Computing it confirms the reasoning: about 0.81 years against the 0.98 years this problem found.)

What to accept in part (c): any argument that compares the size of $p(10-p)$ on the two intervals, or that names $p = 5$ as the maximum growth rate. A numerical answer is not required.

8. Challenge: total emission, $\displaystyle\int_0^\infty \frac{6}{(t+1)(t+4)}\,dt$**.**

Step 1. Part (a): $6 = A(t+4) + B(t+1)$. At $t = -1$, $6 = 3A$ so $A = 2$; at $t = -4$, $6 = -3B$ so $B = -2$. The antiderivative is $2\ln|t+1| - 2\ln|t+4| = 2\ln\left|\frac{t+1}{t+4}\right|$.

Step 2. Part (b): on $[0, b]$ this is $2\ln\frac{b+1}{b+4} - 2\ln\frac{1}{4} = 2\ln\frac{b+1}{b+4} + 2\ln 4$. The quotient tends to 1, so its logarithm tends to 0. $\boxed{2\ln 4}$

Step 3. Part (c): the limit is finite, so the total emission converges. $\boxed{\ln 16 = 2\ln 4}$

Step 4. That is about 2.77 kilograms released over all time, even though the process never stops emitting.

The conceptual half of part (c): decaying to 0 is not enough. The rate $\frac{1}{t+1}$ also decays to 0, yet $\int_0^b \frac{dt}{t+1} = \ln(b+1) \to \infty$ — the total diverges. What matters is *how fast* the rate decays: this problem's integrand behaves like $6/t^2$ for large t, which decays fast enough to give a finite total. Accept $1/t$, $1/(t+1)$, or any $1/t^q$ with $q \leq 1$.